DOST Form 2B

DETAILED RESEARCH & DEVELOPMENT PROJECT PROPOSAL

(For the Component Project)
(To be accomplished by the researcher)

(1) PROJECT PROFILE

Project Title: A Comprehensive PILI (Canarium ovatum Engl.) Information System Utilizing Web 2.0 Technologies: A Roadmap to KM Buildup

Studies: (1.) Design and Development of a Comprehensive Pili Information System Backend
(2.) Design, Development and Deployment of a Pili Information System Portal
(3.) Integration of Data Analytics and E-Commerce Engines
(4.) Capability Building and Evaluation

Leader/Gender: Dr. Ronnel R. Atole / Male - Project Leader
Mr. John Rey S. Lirag - Study 1 Leader (Project Staff Level 2)
Mr. Rowell John B. Artiaga - Study 2 Leader (Project Staff Level 2)
Mr. Leo Constantine S. Bello - Study 3 Leader (Project Staff Level 2)
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(2) COOPERATING AGENCY/IES

BCAARRD
BCAARRD Member Agencies

(3) SITE(S) OF IMPLEMENTATION (Municipality / District / Province / Region)

Region V- Bicol

(4) TYPE OF RESEARCH

_____ Basic
__x__ Applied

(5) R&D PRIORITY AREA & PROGRAM (based on HNRDA 2017-2022)

__x__ Agriculture, Aquatic and Natural Resources
Sector: _____
_____ Health
Sector: _____
_____ Industry, Energy and Emerging Technology
Sector: _____
_____ Disaster Risk Reduction and Climate Change Adaptation
Sector: _____
_____ Basic Research
Sector: _____

(6) RATIONALE (not to exceed one page)

This project, in general, responds to the challenges entailed in the national government's having identified Pili as a regional priority crop of Bicol. In particular, it seeks to address the dearth of accessible knowledge on the Pili industry which could be contributory to its having been a marginalized sector in the country. Thus, the proponents, through this endeavor, envisions a

comprehensive Pili Information System that will be a repository of information not only to BCAARRD and its member-agencies but to Pili farmers/producers, vendors, enthusiasts, retail outlets, and the market as a whole. It will be designed with flexibility in mind to interface and interconnect/link with other PCAARRD systems allowing for a dynamic web of knowledge.

Commercialization of pili products, though to some extent has been a success, is yet to be backed by readily accessible data and knowledge. Though pili products are everywhere these days, there is no comprehensive and easily retrievable knowledge on this particular industry. It is believed that making information and data on Pili publicly available will enliven and invigorate the stakeholders in this industry and help boost yield / production, encourage more research and development efforts, promote innovation, step up marketability of various pili-related products, and consequently improve the living conditions of marginalized farmers and vendors.

Awareness on the part of farmers and vendors, on the marketability and viability of investing into the pili industry will redound to a more vibrant area of this agricultural sector. Moreover, scientists and researchers will be given access to a comprehensive source of information about Pili ranging from its classifications, production sites, uses/applications, products, production data, plantations, technologies, profile of farmers, profile of vendors and suppliers, prevailing market conditions, and many other information.

The integration of Web 2.0 technologies into the design concept of Pili Information System proposed will open opportunities for collaborative access to the knowledge base similar to social media interactions and virtual communities. Farmers, vendors, researchers, other stakeholders and the community as a whole will be able to contribute contents into the information repository through their accounts, creating a participatory environment. This collaborative environment is known to provide a culture of motivation and spark interests among key players in the cultivation, production and commercialization to adapt and move to the pace of what the world market needs.

(7) REVIEW OF LITERATURE (not to exceed three pages)

a. Agricultural Knowledge Management Systems

Numerous studies have been conducted applying Information and Communication Technologies (ICT) in many sectors of the society. In Agriculture, there are those aimed to exploit ICT's power in managing, organizing and disseminating information. As early as 1998, the International Rice Research Institute (IRRI) had the International Crop Information System (McLaren, et.al., 1998), a database system for the management and integration of global information on genetic resources and crop improvement of any crop. It offered comprehensive crop information system that is implemented separately in each crop and is based on unique identification of germplasm, management of nomenclature and retention of all pedigree information. From this system, the International Rice Information System (IRIS) was developed showing promising prospects.

A more recent one is the Mango Resource Information System (Rajan, et.al. , 2013) which was developed for the management of the phenotypic, genetic, molecular, chemical and other available information on indigenous and exotic mango cultivars. The information system is enriched by the collection of various mango related information ranging from its accession, genomic sequences, to protein information present in different cultivars. Published information links, including papers, reports, proceedings, journals from 1972 to 2008 are present in its reference section along with other features such as discussion forum. It featured fast access to information and on extracting specific information from large data sets.

In 2010, the European Union funded a project called FutureFarm (C.G. Sorensen, 2010). It springs from a conceptual modelling and prototyping of a new Farm Management Information System designed to handle adaptation to the shifting paradigm of European farmers. The information system was used to advise managers, recommend guidelines and documentation requirements for various decision making processes, in order to effectively meet farmers' changing needs.

Weather variability as it affects the agriculture sector, had also been subject of Information System development efforts. AgClimate, a climate forecast information system (Fraisse, et.al., 2006) is an ICT tool that responds to the needs for information on proactive adaptation to seasonal and inter-annual

climate variability forecasts in southern USA. The system benefits extension agents, agricultural producers, forest managers, crop consultants and policy makers in the decision making that might mitigate potential loses as well as those with potential to produce optimal yield. AgClimate is a web-based climate forecast and information system. It is composed of two (2) components: the font-end interface, and a set of dynamic tools. It features comprehensive information on climate and El Niño, a tool that allows users to examine climate forecasts for individual countries based on ENSO phase and an evaluation tool of the crop's potential yield.

b. Information Systems in Other Related Sectors

The Environmental Information Systems (EIS) in Sub-Saharan Africa (Prepost, 1999) was developed with the advent of natural resources and environmental action plans. The system covered data, statistics, and other documents that enabled managers to identify and quantify specific environmental categories and to determine their optimum utilization. EIS includes strategies, procedures and institutional framework, together with data management tools that ensure access to relevant data and allow their analysis. It is designed to support wide variety of users and assist them in their decision. The EIS encompasses several technologies specially a Geographic Information System (GIS) which serves to combine geo-referenced information on natural resources and other essential data such as demographics, to support planning, management and decision making.

FishBase is a global biological database on fish that is developed by the International Center for Living and Aquatic Resource Management. Reports claim it contains more than 6,000 species, 8,600 synonyms and about 2,000 common names; population dynamics for about 1,000 species and genetic information for about 300 species. The database is reproduced in CD-ROM and runs in IBM compatible computers. Various cooperating agencies used FishBase as backbone of new information systems, and contributed on the further development of the said database.

c. ICT and Agricultural Knowledge Management

ICT plays an important role in the advancement of Knowledge Management strategies. A report from United Nations Development Program (UNDP) support such claims (Ewalya, 2011). For instance, in Ethiopia, there is a Growth and Transformation Plan (GTP) to improve rural livelihood and eradicate poverty, among other purposes. As part of the GTP, utilization of knowledge management enhanced the agricultural productivity. Knowledge management enabled appropriate knowledge and information to reach intermediate and smallholder farmers in a timely manner. Sources of Agricultural Knowledge include scientific and indigenous knowledge and in the transfer of these to the farmers, ICT facilitated rapid, effective and cost efficient knowledge management. Effective knowledge and information management in the agricultural sector as the report suggests, will be achieved when the right knowledge and information is delivered to the farmer and other stakeholders at the right time in a user friendly and accessible manner.

China also leverages on the use of ICT through various information dissemination models (Zhang, et.al., 2016) to farmers and farm commodities to improve productivity and economic, social and environmental sustainability. The models range from Web Portals, Voice-Based service, Text-Based Services, Online Community, Video Conferencing, Mobile Internet-Based Services to multi-Channel service model. Most ICT-based models have been established in 2000's. Upon the review of the models, it is found out that through ICT-based information and dissemination services, farmers are able to improve income and economic situation through better practices though challenges still exists such as poor IT infrastructure and internet coverage in remote areas, lack of technical support staff, limited information sources, and lack of rigorous information and quality control.

d. Web 2.0

Information and Communication Technologies (ICTs) have greatly influenced various activities though improved communication and access to information (Jayade, et.al., 2013). Internet Technologies as part of the evolution of the communication have adapted to the needs of its stakeholders or users. From Web 1.0 or the "web of cognition" - the age of static webpages wherein information transfer is unidirectional, web pages running under Web 2.0, 3.0, and 4.0 technologies are now the norm. Major features of Web 2.0 include social networking functionalities, user-created websites, self-publishing

platforms, tagging and social book marking. Users can provide data/ share content on a Web 2.0 site and exercise some control over the information.

Web 2.0 sites may likewise allow users to interact and collaborate with each other in a social media dialogue as creators of user generated content in a virtual community. Some of the examples of the web 2.0 include social networking sites, blog, wikis, video sharing sites, hosted services and web applications that allow users to do more than just retrieve information. They provide users with user interfaces, social and storage facilities in the browser. Among the commonly used Web 2.0 tools are Facebook, Wikis, Blogs, Google Docs, Flickr and Twitter. These tools carry the potential of complementing, improving and adding collaborative dimensions to the many web-based services (Tambouris, E., & Tarabanis K., 2012).

Web 2.0 enhances web-based collaborative-authoring or content management thus facilitating creating and editing contents (Salajan & Mount, 2012). These technologies allow collaborative writing, content sharing, social networking, social bookmarking and syndication (O'Reilly, 2007). It offers potential for knowledge management. Web 2.0 is suitable for electronic research, knowledge creation, sharing, storing and usage (Mtega, et.al., 2014). It provides a suitable platform for sharing innovations and electronic learning.

In delivering and sharing agricultural information, the study of (Mtega, et.al. , 2014) assessed the awareness and use of Web 2.0 as a tool for disseminating agricultural knowledge in Tanzania. It defined Web 2.0 tools as user-centered, allowing high degree of customization, allowing users to quickly and easily add and remove content, contribute to the application's content and facilitate social networking possibilities.

(8) OBJECTIVES

General Objectives:

The proposed project aims to develop a comprehensive information system for Pili crop utilizing Web 2.0 technologies. Specifically, the project has the following objectives in mind:

Specific Objectives:

i. Design and develop a comprehensive Pili Information System backend covering logical and physical database components;
ii. Design, develop and implement the corresponding Pili Information System frontend incorporating collaborative social network functionalities, visual features, reports generation, and other components;
iii. Deploy the Information System in a public IP address and domain name;
iv. Integrate Data Analytics and E-Commerce engines into the Information System and interface modules;
v. Establish guidelines, policies, protocols, procedures for the maintenance and operation of the system;
vi. Conduct capability building activities/trainings on the stakeholders and BCAARRD staff; and
vii. Evaluate the Pili information system based on ISO 25010:2011 quality model.

(9) METHODOLOGY (See guide at the back for details)

a. Data Gathering
Secondary data gathering will be conducted to compile knowledge on Pili ranging from production data, volume of sales, profile of farmers and vendors, biological characteristics of Pili and diversities, etc. Research data and outputs on Pili, as well as records will also be considered. These will form part of the knowledge base repository to be housed in the KM solution.

During the evaluation phase, a survey instrument will be developed respondents of which will represent all stakeholders in the Pili industry such as vendors, farmers, manufacturer, researchers, consumers, etc. For this purpose, a simple descriptive treatment of data and qualitative approach will suffice.

In other situations, key-informant interviews and focused group discussions will be conducted such as in formulating guidelines, policies, and procedures for the Information System operation, or in validating secondary data. These may also be resorted to, in gathering inputs from stakeholders regarding the performance of the prototype Information System in terms of the parameters under ISO 20510:2011 standards as shown in the Conceptual Framework.

b. IS Development Process

A simplified view of IS development process shall be followed, eliminating such details intimated in models like Waterfall, Spiral, etc. This simplified view, based on our experience, suffices, eliminating rigors that are only required in academic purposes.

i. Requirements Analyses and Scoping

Considered by many as the most important phase in IS development, requirements analyses and scope definition involve identifying the data, functional, and other requirements that the system solution must provide. In the context of the Pili Information System proposed, this includes characterizing what information the system must be able to handle and provide, what functionalities the web portal should provide, security mechanisms to put in place, and the like. Structured interviews, secondary data gathering, sampling of form and records are some of the techniques that will be used.

ii. Design

Main KM Architecture

The proposed system will be designed to be a framework of knowledge sharing and retrieval among various entities responsible for promoting, cultivating and managing Pili information.

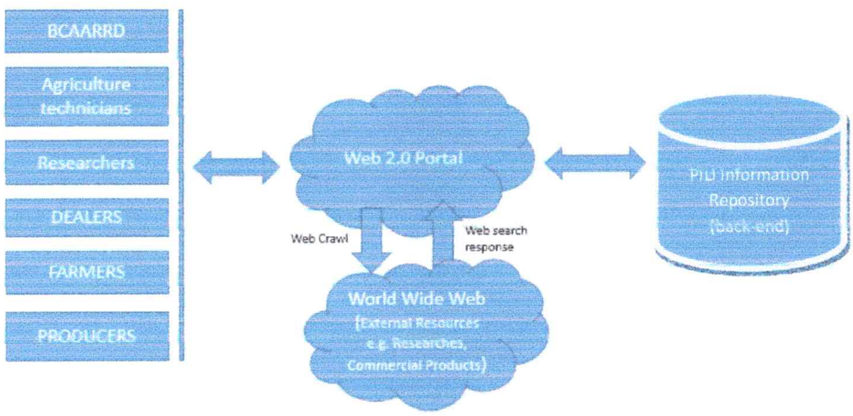


Figure 3. Architectural design of the Pili Information System

The Web 2.0 portal serves as an interface in facilitating access of information stored in the Pili KM database. The system will be deployed in the Web for instantaneous and wide accessibility of information from any location.

The users of the system will be divided into different roles, namely:

1. Administrator. The administrator is responsible for the maintenance of the system. This entity is responsible for configuring users' access as well as the credentials of the users.
2. Content Review. This entity moderates and checks information for content validity before it will be published in the portal. They ensure that the information being posted is reliable and correct.
3. Contributors. These entities will be responsible for the buildup of the knowledge related to Pili. Government Agencies on Agriculture can be considered as contributors, Agriculture Technician, Academes and Agriculture researchers can be considered in this role. These entities are also responsible for the information update on Pili Human resource – Growers, Vendors, Processors and Traders.
4. Viewers. These entities are commonly the end users. They can view information stored in the database and seek Pili related information but their inputs are limited to only in participation in forum discussion.

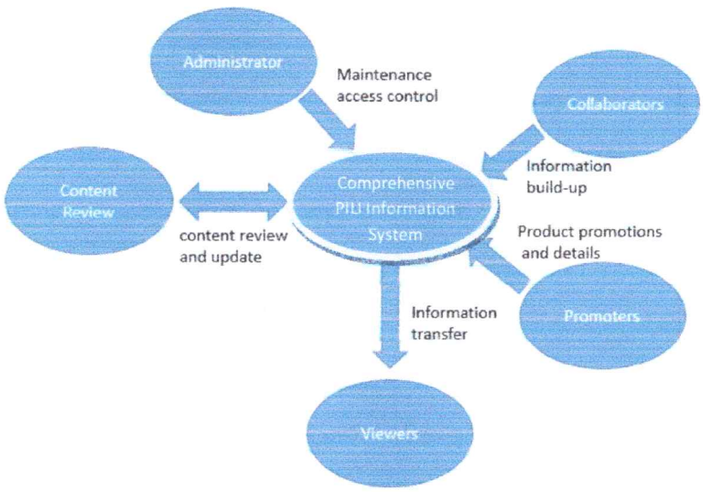


Figure 4. Pili Information System entities and their respective roles

Database Design

One of the objectives is to design and develop a repository that will handle massive Pili related data. The database will be normalized to handle complex queries needed to generate knowledge. Pili related information will be categorized into 4, namely:

1. Base information. The base information will be comprised of the Taxonomy, Botanical Description, Properties, Molecular Biology, Habitat and Ecology, and Cultivation and Uses. This will include all information needed to understand Pili when it comes to its physical properties and uses.

2. Promotions. This will include information needed to commercialize and promote pili products and by-products.
3. Human Resources. Information on entities such Pili growers, vendors, traders, processors alongside with their location and contact information and their products. Information on agencies is also being sought in this category.
4. Research and Development. This will be comprised of information on Pili related studies and researches.

iii. Coding / Implementation

Coding or implementation refers to the process of converting the design into actual system solution. All the requirements identified in the Requirements Analyses phase will be delivered through integration. Various ICT components will be integrated to develop the KM system, such as the following:

1. Database and warehousing technologies. These technologies will form the repository of data and knowledge and together with the frontend system.
2. Internet technology is needed for the instantaneous communication between key stakeholders of the system. Most of the time the system will rely on this technology for linking various resources and facilitate the transfer of information from its server to the clients, and vice versa (content update and discussion).
3. Web 2.0 technology. To build the system in a cost-effective manner, the project will utilize the power of the Web 2.0 technology specially its participatory collaborative feature in building up knowledge.
4. Web mining techniques can be used in extracting external resources from the ambiguous information scattered across the internet. This can be used to link other related information especially those that are not present in the current database.

iv. Testing

Testing is necessary to see to it that the design objectives are carried out and root our potential errors. This phase will include both unit and system wide tests and will involve all prospective stakeholders and users. Their feedback and inputs will be used to improve, optimize and enhance the system. A possible approach is the conduct of interviews and focused group discussions as well as rapid prototyping in which system prototypes will be presented to users for inputs, in an iterative process.

v. Deployment and Operation

Once deployed, users can sign-up their respective credentials to the page. These user information will be validated by the administrator before giving them access to the portal. After approval, they will be given username and password which will be used to log-in to the portal.

User views and accesses to the system's tools will vary depending on the role being given by the administrator. Contributors will be given access to Information Tools and Promoters will be given access to Promotions Tools. Users without credentials could visit the site for Pili related information like access to the Taxonomy and Biological

Description on various Pili species, etc. but is limited only to viewing and not in the update of the information. Viewers can participate in discussion related to the Pili Information. Content review views all content and checks for possible violations before posting it to the web for public view.

(10) TARGET ACCOMPLISHMENTS AND EXPECTED OUTPUTS

Tasks	Expected Output
Data gathering and consultation	Project scope, comprehensive data on Pili crop such as genetic characteristics, profile of farmers/ vendors/ dealers/ experts, market data on Pili-related products, etc.
Requirements Analysis (Data and Physical Database)	Characterized data requirements
Procurement of database platform	Database Server made available
Database design and normalization	3rd degree normalized database
Physical database development	Physical database configured and installed
Implementation and testing	Optimized database Security features included
Progress Reports, and other Reports (Conference Presentations and Publication of Outputs)	Reports, publications and conference presentations
Information System Requirements Analyses (Software and Hardware)	Characterized software and hardware requirements
Procurement of Software for IS Frontend	Development platform for IS front-end made available
Designing and coding	Website prototype
Prototype testing and presentation	Comments, recommendations, final User Interface design
Finalization and deployment	Delivered and operational Pili Information system
Design and modeling of Data Analytics Components	Designed data analytics tools
Implementation of Data Analytics Components	Implemented tools for data analytics with visualizations such as graphs, GIS, etc.
Integration of e-Commerce features	Integrated e-commerce module on the Pili Knowledge Management system
Trainings of BCAARD staff	BCAARRD personnel and staff acquainted and familiarized with the KM operational procedures
Trainings of Stakeholders (i.e. Farmers, Researchers, BCAARD KM Coordinators & Staff, Vendors, etc.)	Farmers, researchers, vendors and stakeholders familiarized with the KM operational procedures particularly on content uploading.
Evaluation of the Pili Information System based on ISO 25010:2011 quality model	Performance and/or quality of the Information System characterized
Submission of Reports	Required reports duly submitted

(11) TARGET BENEFICIARIES

PCAARD
PILI Experts/Researchers
PILI Growers, Farmers, Producers, Vendors
PILI Knowledge Contributors

(12) GENDER AND DEVELOPMENT (GAD) SCORE (based on the Harmonized Gender and Development Guidelines)

2 – Installation of Strategic Mechanism

(13) LIMITATIONS OF THE PROJECT

The project will focus on the design and development of an information system about Pili (Canarium Ovatum Engl.) being the recognized flagship crop in the Bicol Region. Part of the project is the initial content buildup geared to support KM requirements. The system will include physical and logical design of the Pili database (back-end), interactive front-end, e-commerce functionalities as well as Data Analytics features.

The project also includes capacity building on the part of prospective stakeholders and users, as well as evaluation based on the parameters in ISO 25010:2011 quality model. It will, however, exclude impact assessment, procurement and installation of hardware components such as network, servers, and facilities.

(14) LITERATURE CITED

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(15) PERSONNEL REQUIREMENT

Position	Percent Time Devoted to the Project	Responsibilities
Program/ Project Leader: Ronnel R. Atole, Ph.D.	**50%**	Lead overall development of the project
Study Leaders:		
Mr. John Rey S. Lirag	**40%**	Lead the design and development of Pili Information System Repository (Back-end)
Rowell John B. Artiaga	**40%**	Lead the design and development of the Pili IS front-end
Leo Constantine S. Bello	**40%**	Lead the design and integration of the e-commerce and data analytics functionality of the Pili IS
Salvador V. Briones, II	**40%**	Lead the capacity building and impact assessment of the project
Training Resource Person (Lecture)	**5%**	
Training Resource Person (Laboratory)	**5%**	
Training Support Staff	**5%**	
Training Support Staff	**5%**	

(16) BUDGET BY SITE OF IMPLEMENTATION

Site of Implementation	PS	MOOE	EO	Total
Year 1				
Site 1	285,600.00	176,920.00	261,040.00	723,560.00
Year 2				
Site 1	248,880.00	210,681.00	0	459,561.00
Year 3				
Site 1				
TOTAL	534,480.00	387,601.00	261,040.00	1,183,121.00

(17) OTHER ONGOING PROJECTS BEING HANDLED BY THE PROJECT LEADER: _____ (number)

Title of the Project	Funding Agency	Involvement in the Project
NONE		

I hereby certify the truth of the foregoing. Any willful omission/false statement shall be a basis of disapproval and cancellation of the project.

(18)	SUBMITTED BY (Project Leader)	ENDORSED BY (Head of the Agency)
Signature		
Printed Name	Ronnel R. Atole, Ph.D.	RAUL G. BRADECINA, Ph.D.
Designation/Title	Director of ICTMO	SUC President III
Date		

Note: If the project is part of a program, this form should be submitted together with detailed R&D proposals for the whole program. See guidelines/definitions at the back. This form can be used for Technology Transfer Projects.